



are fatigued, distracted, or otherwise impaired. This makes driver monitoring a compelling example of AI-enabled functionality in vehicles.

Beyond safety features, AI is also used by automotive manufacturers to personalise the in-cabin experience. For instance, AI can identify the driver through facial recognition and automatically adjust the seat position, mirror angles, and other settings to match their preferences. These customisations are made possible by combining AI with in-cabin sensing hardware such as cameras and sensors.

There is significant potential to further integrate in-cabin sensors with AI, particularly in the context of software-defined vehicles (SDVs) and connected cars. As specific hardware components are already mandated by regulation, automotive OEMs can enhance their value by adding software features. This approach allows them to offer added benefits to drivers and potentially monetise these features as part of their service offerings.

To ensure the safety and reliability of AI systems in vehicles, regulatory standards and validation procedures are in place to ensure their safety and reliability. For instance, in Europe, organisations such as Euro NCAP evaluate the performance of driver monitoring systems, including the AI models that power them. Automotive manufacturers also perform internal testing to assess the robustness and reliability of these systems.

Sensors meet software

In-cabin sensing and SDVs are not just related—they are evolving together. Regulatory requirements are prompting automakers to incorporate hardware, such as near-infrared cameras, for driver monitoring. Once installed, this hardware enables a variety of value-added features that OEMs can monetise.

A common feature is automatic

Regional Trends

DMS show global alignment, with near-infrared (NIR) cameras being the standard across regions. However, OMS vary by market:

- **North America:** Radar sensors are widely adopted.
- **China:** Time-of-flight cameras are commonly used, although radar is also employed.
- **Europe:** The focus is on data privacy, especially regarding GDPR compliance, with strong attention to whether data is stored locally in the vehicle or uploaded to the cloud. While data protection matters globally, Europe's regulations are generally stricter.

driver identification. When one enters the car, the system can recognise the person and automatically apply the person's personalised settings, seat position, mirrors, radio stations, and climate preferences. This enhances the user experience and can be sold as a premium feature, aligning with the SDV model in which software upgrades create ongoing revenue.

In-cabin sensing also supports more advanced, AI-driven services. For instance, if the car detects signs of driver fatigue, it could suggest or even pre-order a coffee at the next drive-through location, merging convenience with safety. Companies such as LG are already showcasing systems that combine driver monitoring with advanced driver assistance systems (ADAS) to understand better driver behaviour, including where they are looking on the road.

When cars drive themselves

At present, most vehicles operate at SAE Level 2 or 2.5, with limited adoption of Level 3. At these levels, the driver remains responsible for the car and must stay alert and ready to intervene. Consequently, in-cabin sensing, particularly driver monitoring systems, is essential. Regulations require drivers to remain attentive, so technologies such as eye tracking and drowsiness detection are crucial for safety.

Looking ahead, as vehicles progress to SAE Level 3 and beyond, the function of in-cabin sensing may evolve. At these higher levels,

drivers are no longer required to monitor the road at all times; the vehicle assumes full control, allowing the human to disengage. In such scenarios, tracking the driver's attention or fatigue may become unnecessary.

However, OMS will remain important. Even in fully autonomous vehicles, it will be essential to detect who is in the car, monitor vital signs such as heart rate, and provide personalised settings for comfort and convenience. While driver monitoring may diminish over time, occupant sensing is expected to remain a core feature of future vehicles.

In-cabin sensing is rapidly becoming a cornerstone of both current and next-generation vehicles. Originally designed to ensure driver awareness and passenger safety, it has evolved into an advanced suite of sensors and AI-driven software that enables personalisation, meets regulatory demands, and opens up new opportunities for innovation and revenue. With an understanding of how these technologies function and what they can enable, readers are better equipped to assess, contribute to, or advocate for more innovative, safer, and more intuitive automotive experiences. Whether involved in the industry or simply passionate about the future of mobility, this knowledge provides a meaningful advantage. Take the next step: stay informed, get involved, and help shape how we interact with vehicles going forward. **EFY**

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